

FuzzyContact: A generic contact mechanics formulation via a modified penalty method governed by sigmoidal traction law

Fatih Uzun^{*}, Alexander M. Korsunsky¹

Department of Engineering Science, University of Oxford, Oxford, UK

ARTICLE INFO

Keywords:

Computational contact mechanics
Proximity-based contact
Stick-slip friction
Finite element method (FEM)
Neutron diffraction

ABSTRACT

Conventional contact algorithms frequently suffer from numerical instabilities and mesh sensitivity due to their reliance on discontinuous, binary logic to enforce boundary constraints. This paper introduces FuzzyContact, a generic contact mechanics formulation that implicitly captures contact interaction phenomena within a unified, continuous framework. The method employs a sigmoidal traction law that defines contact forces based on inter-nodal proximity, eliminating the need for predefined master-slave surfaces or contact stiffness matrix assembly. This proximity-based approach utilizes a modified penalty parameter in the units of force to decouple interaction strength from mesh-dependent penetration length, ensuring a smooth, differentiable transition between stick and slip states governed by dynamic relaxation. The accuracy and versatility of the formulation were validated through a series of benchmarks, culminating in a direct comparison with experimental data. Simulated elastic strains in a compressed steel cylinder showed excellent agreement with measurements from *in situ* neutron diffraction. The proposed framework provides a robust and scalable solution for generic contact simulations, paving the way for advanced modelling of complex tribological systems involving friction and wear.

1. Introduction

Accurate modelling of contact interactions has long represented a central challenge in computational mechanics and the broader field of tribology, beginning with the classical analytical solutions of Flamant and Boussinesq for point loading on a semi-infinite solid and culminating in Hertz's seminal 19th-century work on smooth elastic contact, which provided the first continuum solutions for pressure distribution and deformation [1]. Subsequent developments extended Hertzian theory to include adhesion through the Johnson-Kendall-Roberts (JKR) [2] and Derjaguin-Muller-Toporov (DMT) [3,4] models, while the Greenwood-Williamson (GW) approach [5] introduced statistical treatments for surface roughness, describing asperity interactions in real contacts. Modern computational methods such as the Finite Element Method (FEM) [6–15], Boundary Element Method (BEM) [16–20], Discrete Element Method (DEM) [21–24], and Molecular Dynamics (MD) [25] have since enabled detailed analyses of nonlinear and multiscale contact phenomena. Despite these advances, major challenges remain in addressing high computational cost, mesh dependency, and parameter sensitivity, particularly for irregular geometries and

nonlinear materials.

Within finite element analysis, the penalty method is widely adopted to enforce contact constraints by applying forces proportional to inter-penetration depth [26–31]. The method's success depends on selecting an appropriate penalty stiffness [32–35] because insufficient stiffness permits unrealistic penetration, whereas excessive stiffness causes ill-conditioning of the global system and loss of numerical stability [36–39]. Penalty parameters are strongly influenced by mesh resolution and material stiffness [40,41], and the performance of geometric contact-search algorithms also depends on consistent parameter scaling. While many traditional formulations employ constant penalty stiffness, such simplifications can compromise both accuracy and convergence. Nonlinear and micromechanical models, including the GW and Cooper-Mikic-Yovanovich (CMY) framework and thermomechanical contact formulations based on microscopic interface laws [42], improve realism by defining nonlinear contact pressure-gap relations that account for surface roughness and multi-asperity behaviour. These have been extended to incorporate coupled thermomechanical effects [43], linking contact stiffness to thermal and electrical conductance [44]. Power-law-based contact relations, derived from fractal descriptions of

^{*} Corresponding author.

E-mail addresses: fatihuzun@me.com, fatih.uzun@eng.ox.ac.uk (F. Uzun), alexander.korsunsky@gmail.com (A.M. Korsunsky).

¹ Current affiliation: Center for Digital Engineering, 30 Bolshoi boulevard, Moscow 121205

surface roughness, provide physically meaningful stiffness evolution and have inspired multiscale FEM-BEM coupling strategies [45], enabling local traction evaluation at Gauss-point level. Moreover, compressive power-law formulations share a close theoretical connection with cohesive and adhesive traction-separation laws used for interface elements [46,47], facilitating unified treatment of tensile and compressive contact.

Mortar methods represent another major class of contact algorithms [48,49], enforcing interface constraints between non-matching meshes through variationally consistent Lagrange multiplier formulations [50]. Segment-to-segment mortar strategies, such as those developed by Puso and Laursen, mitigate the force oscillations typical of node-to-segment methods [51] and produce smooth force transfer during large sliding [50]. Despite their theoretical rigour, mortar techniques require complex surface integration over curved geometries and careful quadrature near edges to avoid spurious stress oscillations [52]. Combined penalty-mortar or augmented Lagrange implementations [53,54] improve frictional contact modelling but may still exhibit mesh locking for stiff or highly refined interfaces. Although mortar methods successfully handle non-matching discretizations [55], their implementation complexity, reliance on predefined contact pairs, and integration overhead remain practical challenges.

Asymptotic methods provide analytical insight into local contact phenomena by resolving stress singularities near corners and edges [56]. These formulations employ eigenfunction expansions [57,58] to capture power-law traction distributions that characterize contact edges and crack-tip fields. Typical solutions involve two asymptotic eigenfunctions [59,60] representing radial and angular variations of stress near the contact boundary. For linear elasticity, the absence of an intrinsic length scale allows separation of variables and characterization of stress fields purely through geometry and boundary conditions. Such approaches have also been used to predict crack initiation in fretting-fatigue and edge-contact problems [61,62], illustrating their versatility for localized failure analysis.

While power-law asymptotics effectively describe stress singularities, smooth functional forms are required to represent contact interactions across finite regions. Smooth decay functions [63], including logistic (inverse-sigmoid), exponential, smooth-step, and Gaussian profiles, offer continuity and differentiability that facilitate numerical implementation. These functions transition gradually from unity at zero gap to zero at large separation and are increasingly applied in modelling contact stiffness [64–66], frictional response [67], and adhesive interactions [68–71]. Their continuous nature provides stable convergence and realistic transitions in stress, deformation, and energy dissipation, making them well suited for contact formulations that integrate geometric proximity with nonlinear traction laws.

The development of FuzzyContact is motivated by the inherent limitations of conventional surface-based contact algorithms, which rely on predefined master-slave surfaces and discrete tracking procedures. The proposed formulation replaces point-to-segment detection with a continuous proximity field, where the nodal contact force is a weighted function of distances to neighbouring nodes. Built within a modified penalty framework, a sigmoidal traction law captures an implicit stick-slip response through history-dependent proximity updates, achieving continuous force evolution without conditional logic or separate friction subroutines. This unified scheme, which utilizes a penalty parameter in units of force rather than stiffness, enhances numerical robustness and ensures symmetric treatment of contacting bodies.

To validate the fidelity and robustness of this formulation, a comprehensive suite of benchmarks is presented. This includes a patch test verified against Hertzian theory, simulations of combined normal and tangential loading, and a V-indenter problem cross-validated with a commercial finite element solver. Furthermore, experimental verification is provided through a comparison of simulated elastic strains with in-situ neutron diffraction measurements from a compressed steel cylinder. Finally, a multi-body connected assembly problem is analyzed to

verify mesh convergence and scalability. These benchmarks collectively demonstrate the formulation's accuracy and confirm its predictive fidelity.

2. Methodology

The proposed FuzzyContact method is designed to efficiently and robustly simulate generic contact phenomena. Within this paper, the term 'fuzzy', which here denotes spatial continuity rather than uncertainty, is specifically defined to describe the replacement of sharp, binary (on/off) contact conditions with a continuous and spatially-distributed transition zone, where contact forces evolve smoothly according to a weighted field of geometric proximity. This concept, which refers directly to the gradual and continuous nature of the contact enforcement, should not be confused with formal fuzzy logic theory. This spatial fuzziness is governed by a sigmoidal traction law that eliminates the need for a priori surface identification or master-slave assignments, thereby enhancing numerical robustness and stability.

Unlike traditional master-slave formulations, the proposed approach is inherently symmetric, independent of which body is designated as master or slave. It defines contact interactions based solely on the geometric proximity of all potential nodal pairs, with the sigmoidal traction law activating only when nodes lie within a prescribed proximity range. This represents a key distinction from adhesive models that employ interaction potentials to generate tensile forces. In contrast, FuzzyContact operates purely in compression, and its history-dependent formulation ensures that contact forces evolve consistently with the instantaneous geometric configuration and remain aligned with the surface normal, preserving the physical constraints of non-adhesive contact while maintaining symmetry and robustness.

2.1. FuzzyContact formulation

The core of the proposed formulation is a sigmoidal traction law given by Equation 1, illustrated in Fig. 1, that governs the interaction force based on the axial components of the inter-nodal distance given as d_x , d_y and d_z . This function, which smoothly transitions from zero force to maximum repulsion as nodes approach each other, is defined by three key parameters: the proximity value, a , that activates the force only when the nodal separation is within this range, the transition parameter, b , that sets the point of rapid force escalation, and a power exponent, m , that controls the steepness of this transition. This carefully designed law provides a versatile and computationally efficient method for capturing the complex relationship between nodal proximity and contact force, inherently avoiding the need for master-slave designations.

$$s_x = \frac{(a - d_x)^m}{(a - b)^m + (a - d_x)^m} \quad (1a)$$

$$s_y = \frac{(a - d_y)^m}{(a - b)^m + (a - d_y)^m} \quad (1b)$$

$$s_z = \frac{(a - d_z)^m}{(a - b)^m + (a - d_z)^m} \quad (1c)$$

While the sigmoidal function offers clear mathematical advantages for numerical stability, its selection is also conceptually motivated by the physics of real surface interactions. It is important to clarify that the FuzzyContact formulation does not explicitly resolve individual asperities or model surface roughness statistics. Instead, it employs the sigmoidal law as a phenomenological tool whose behaviour is qualitatively analogous to the homogenized mechanical response of a multi-asperity interface. This analogy can be understood by mapping the stages of physical contact to the function's shape. The initial, low-stiffness toe of the curve is conceptually consistent with the initial contact of only the highest asperities. The steep central region mimics

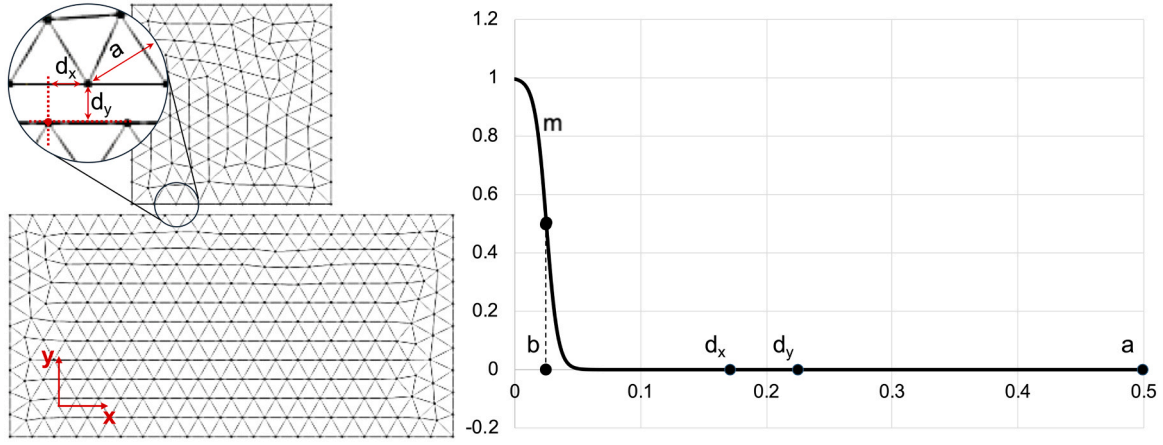


Fig. 1. Schematic of the FuzzyContact interaction between opposing surface nodes, highlighting the parameters of the sigmoidal traction function: proximity range a , transition parameter b , and power exponent m , that govern the magnitude of the contact force based on inter-nodal distance.

the qualitative behaviour of a rapidly increasing population of contacting asperities, which leads to a sharp increase in interface stiffness. Finally, the shoulder of the curve, which asymptotically approaches a hard-contact state, represents the saturation of this process as the interface becomes fully compacted and the bulk material response begins to dominate. Therefore, the sigmoidal traction law provides a physically-plausible and continuous representation for the evolution of interface stiffness, bridging complex micro-scale concepts with a computationally tractable macro-scale formulation.

The implementation of the described traction rule treats each surface node as a potential point of contact. For a given node, the sigmoidal function given in Equation 1 is evaluated for all neighbouring nodes on the opposing body that lie within the proximity range a . This process yields a set of interaction strength values, s_x , s_y and s_z , one for each interacting neighbour. These individual strengths are consolidated into a single, representative measure, the resultant proximity parameters, P_x , P_y and P_z , using the weighted average shown in Equation 2. This equation performs a summation of the individual, s_x , s_y and s_z , values, weighted by their respective proximity. This weighted averaging approach ensures that nodes in closer proximity exert a greater influence on the final parameters, P_x , P_y and P_z , thereby accurately capturing the localized nature of physical contact interactions.

$$P_x = \frac{\sum_i s_{x,i} (a - d_{x,i})}{\sum_i (a - d_{x,i})} \quad (2a)$$

$$P_y = \frac{\sum_i s_{y,i} (a - d_{y,i})}{\sum_i (a - d_{y,i})} \quad (2b)$$

$$P_z = \frac{\sum_i s_{z,i} (a - d_{z,i})}{\sum_i (a - d_{z,i})} \quad (2c)$$

A central task in capturing the mechanical interaction is the determination of the contact force vector at each surface node. This is achieved using Equation 3, which incorporates the local surface normal, $\mathbf{N}(N_x, N_y, N_z)$, and introduces directional penalty parameters, denoted as k_x , k_y , and k_z . It is essential to distinguish the conceptual basis and units of these parameters from those in a traditional penalty method. A traditional approach calculates force as the product of a penalty stiffness (in units of N/m) and a physical interpenetration distance (in units of m), directly linking the repulsive force to a physical displacement. In contrast, the FuzzyContact formulation is fundamentally different, as it uses the dimensionless proximity parameters, P_x , P_y and P_z , as a normalized measure of the interaction strength. Consequently, for the resulting contact force terms, f_x , f_y and f_z , to be dimensionally correct (in

units of N), the penalty parameters, k_x , k_y , and k_z , must themselves have units of force. This parameter is therefore conceptualized as a 'penalty force scale' rather than a 'penalty stiffness'. Physically, it represents a characteristic force magnitude that scales the overall interaction, while the sigmoidal function governs its nonlinear evolution with proximity. This modification is a core and intentional feature of this framework, as it decouples the magnitude of the contact force from a specific, mesh-dependent penetration length, thereby enhancing numerical robustness.

$$f_x = \chi_x(P_x k_x) + \chi_y(P_y k_y) \mu_y + \chi_z(P_z k_z) \mu_z \quad (3a)$$

$$f_y = \chi_x(P_x k_x) \mu_x + \chi_y(P_y k_y) + \chi_z(P_z k_z) \mu_z \quad (3b)$$

$$f_z = \chi_x(P_x k_x) \mu_x + \chi_y(P_y k_y) \mu_y + \chi_z(P_z k_z) \quad (3c)$$

The directional opposition factors, χ_x , χ_y , and χ_z , are introduced to correctly orient the components of the repulsive contact force. Their formulation, defined in Equation 4, achieves two simultaneous objectives: it ensures the reaction force opposes the direction of the applied external force while also scaling the force component based on the local surface normal, $\mathbf{N}(N_x, N_y, N_z)$. This ensures the resulting contact force vector is consistent with both the loading direction and the surface geometry. These factors are defined as a piecewise function. If a given component of the external force is zero, its corresponding factor is also defined as zero to ensure no spurious force is generated. If the external force component is non-zero, the factor computes a directional value based on the force's sign and the local surface normal. In this piecewise function, the $-F_x/|F_x|$ term determines the sign of the opposition force for the x-component, while the $|N_x|$ term projects this force appropriately onto the x-axis. This logic applies analogously to the y- and z-components, establishing the directional basis for the repulsive forces and scaling the normal force and the tangential frictional resistance. It is important to note that these tangential contributions are strictly dissipative. They arise solely in reaction to relative displacement and cannot generate energy or induce spontaneous motion in the absence of external driving forces.

$$\chi_x = -\frac{F_x}{|F_x|} |N_x| \quad (4a)$$

$$\chi_y = -\frac{F_y}{|F_y|} |N_y| \quad (4b)$$

$$\chi_z = -\frac{F_z}{|F_z|} |N_z| \quad (4c)$$

A consistent, history-dependent response is achieved by

incorporating frictional effects through the coupling terms in Equation 3. In this model, the friction coefficients, denoted as μ_x , μ_y and μ_z , are direct, user-specified inputs that do not distinguish between static and kinetic values. Rather than serving as a direct multiplier for the normal force, these coefficients define the asymptotic limit for the tangential resistance. The key innovation of this formulation is its ability to reproduce the macroscopic behaviour of a Coulomb-type stick-slip transition within a single, continuous, and differentiable mathematical framework.

The continuous, coupled nature of the update naturally reproduces the physical stick-slip process. During incipient relative displacement, the coupling terms cause tangential resistance to grow progressively, which is analogous to elastic sticking. As the driving potential increases, the frictional resistance smoothly and asymptotically approaches the theoretical Coulomb friction limit. This smooth saturation of the tangential force serves as a regularised, differentiable approximation of the sharp transition from static to kinetic friction. This behaviour captures the rapid force growth of the stick phase followed by stabilisation during sliding. Furthermore, the incremental, history-dependent update correctly models both the onset of slip as this limit is reached and the transition back to stick when the driving potential is reduced, establishing this mechanism as a central feature of the formulation.

Following the computation of history-dependent nodal contact forces, the global positions of the contacting bodies are updated through an explicit pseudo-time relaxation step, as given by Equation 5. This update to the body's centre of mass coordinates, c_x , c_y and c_z , is derived from Newton's second law, incorporating the aggregation of all external, F_x , F_y and F_z , computed contact forces, f_x , f_y and f_z , the body's mass, M , and the coordinates from the previous step, c'_x , c'_y and c'_z . Here, δt denotes a pseudo-time increment expressed in seconds, which controls the numerical relaxation rate rather than representing a physical time in a dynamic simulation. The damping coefficient, ζ , (in units of kg/s) serves as an artificial numerical damping factor used to suppress oscillations and ensure monotonic convergence toward equilibrium. This update governs only the rigid-body translation where the internal elastic deformation field is solved separately within the finite-element equilibrium formulation. This hybrid treatment allows the algorithm to iteratively correct body positions while maintaining quasi-static equilibrium of the stress field.

$$c_x = c'_x + \frac{(\delta t)^2}{M^2} \left(\sum_i f_{x,i} + F_x \right) (M - \zeta \delta t) \quad (5a)$$

$$c_y = c'_y + \frac{(\delta t)^2}{M^2} \left(\sum_i f_{y,i} + F_y \right) (M - \zeta \delta t) \quad (5b)$$

$$c_z = c'_z + \frac{(\delta t)^2}{M^2} \left(\sum_i f_{z,i} + F_z \right) (M - \zeta \delta t) \quad (5c)$$

It is critical to note that while Equation 5 adopts the mathematical form of an explicit dynamic update, it functions purely as a numerical relaxation scheme to enforce a quasi-static solution. The static equilibrium problem is recast as a fictitious, heavily damped dynamic system where true inertial effects are rendered negligible. This is achieved by using a sufficiently small pseudo-time increment, δt , which prevents the accumulation of significant artificial momentum from the force updates. The damping coefficient, ζ , serves a purely numerical purpose and is tuned to approximate critical damping for the system's rigid-body modes, ensuring the most efficient convergence by dissipating artificial kinetic energy and eliminating non-physical oscillations without slowing the relaxation process. Each iteration is therefore a pseudo-time relaxation step, not a physical time increment. The system evolves until the net force residual falls below a prescribed tolerance, at which point it is considered to have reached a stable static equilibrium. This iterative process correctly models the physical transition from a stick condition to

slip. If the tangential forces reach the frictional limit, subsequent relaxation steps represent the physical sliding of the body until a new force equilibrium is achieved. This treatment places the overall formulation firmly within the quasi-static regime, enabling robust and efficient convergence.

The computed contact force vector at each surface node is translated into a distribution of surface tractions using Equation 6. This conversion is accomplished by dividing the x -, y -, and z -components of the nodal force by the effective area, A , occupied by that node, ensuring the forces are appropriately distributed for accurate stress state modelling. For computational efficiency, particularly with homogeneous tetrahedral elements, this effective area is approximated as the body's total surface area divided by its number of surface nodes. While this simplification may introduce localized deviations from the true nodal area, it provides a reasonable and efficient approximation of the traction distribution, especially for relatively uniform meshes.

$$T_x = \frac{f_x}{A} \quad (6a)$$

$$T_y = \frac{f_y}{A} \quad (6b)$$

$$T_z = \frac{f_z}{A} \quad (6c)$$

2.2. Quasi-static finite element framework via dynamic relaxation

The mechanical equilibrium of each contacting body is enforced through a variational formulation of the linear elasticity equations, evaluated under quasi-static conditions. Although the contact update employs an explicit, acceleration-based integration form, the purpose of this step is not to capture physical dynamics but to drive the system toward static equilibrium through a dynamic-relaxation procedure. In this approach, inertial terms serve only as numerical stabilizers, while artificial damping removes oscillations and dissipates non-physical kinetic energy introduced during contact-force updates. Each pseudo-time increment thus represents a relaxation iteration within the equilibrium search, ensuring that the solution at convergence corresponds to a static configuration satisfying the balance of internal and external forces.

$$\int_{\Omega} \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega = \int_{\partial\Omega} \mathbf{T} \bullet \mathbf{v} d\Omega \quad (7)$$

Specifically, the weak form of the equilibrium equation is derived from the principle of virtual work, which states that the internal virtual work done by stresses must balance the external virtual work due to applied loads and contact forces for any admissible variation in the displacement field. Mathematically, this principle is expressed in Eq. 7 as for all $\mathbf{v} \in V$, where $\boldsymbol{\sigma}$ is the Cauchy stress tensor, $\boldsymbol{\varepsilon}(\mathbf{v})$ is the strain tensor derived from the test function $\mathbf{v}$, $\mathbf{T}$ represents the surface tractions, Ω denotes the domain of the part, and $d\Omega$ represents its boundary. The contact interaction is incorporated into this formulation through the surface traction term, where contact forces computed using the Fuzzy-Contact formulation, are applied as external loads. The resulting variational problem is then discretized using a finite element basis using an open-source finite element library, and the system of equations is solved using either a direct (Multifrontal Massively Parallel Solver) or iterative solver (Conjugate Gradient), subject to Dirichlet boundary conditions that constrain the displacement field at specific locations. The variational form expressed in Eq. 7 encapsulates the mechanical equilibrium condition by balancing internal virtual work against external virtual work, where the latter includes both prescribed surface tractions and contact forces derived from the FuzzyContact formulation. These tractions are applied as Neumann-type boundary conditions, while essential Dirichlet constraints are imposed to fix displacements at designated regions, ensuring a consistent and well-posed quasi-static formulation. This approach provided a robust and accurate

representation of the mechanical behaviour of the contacting bodies, capturing the interplay between elastic deformation, external loading, and the contact interaction governed by the novel contact formulation.

The contact formulation operates within a hybrid quasi-static finite element framework that decouples internal elastic deformation from global rigid-body motion. The complete computational procedure inte-

grating the FuzzyContact formulation within this framework is illustrated in the flowchart in Fig. 2. A defining feature of this approach is that it circumvents the need to assemble a contact element stiffness matrix where instead, contact interactions are treated as externally applied loads. The rigid-body motion is updated outside the finite element solver using the explicit time-integration scheme derived from

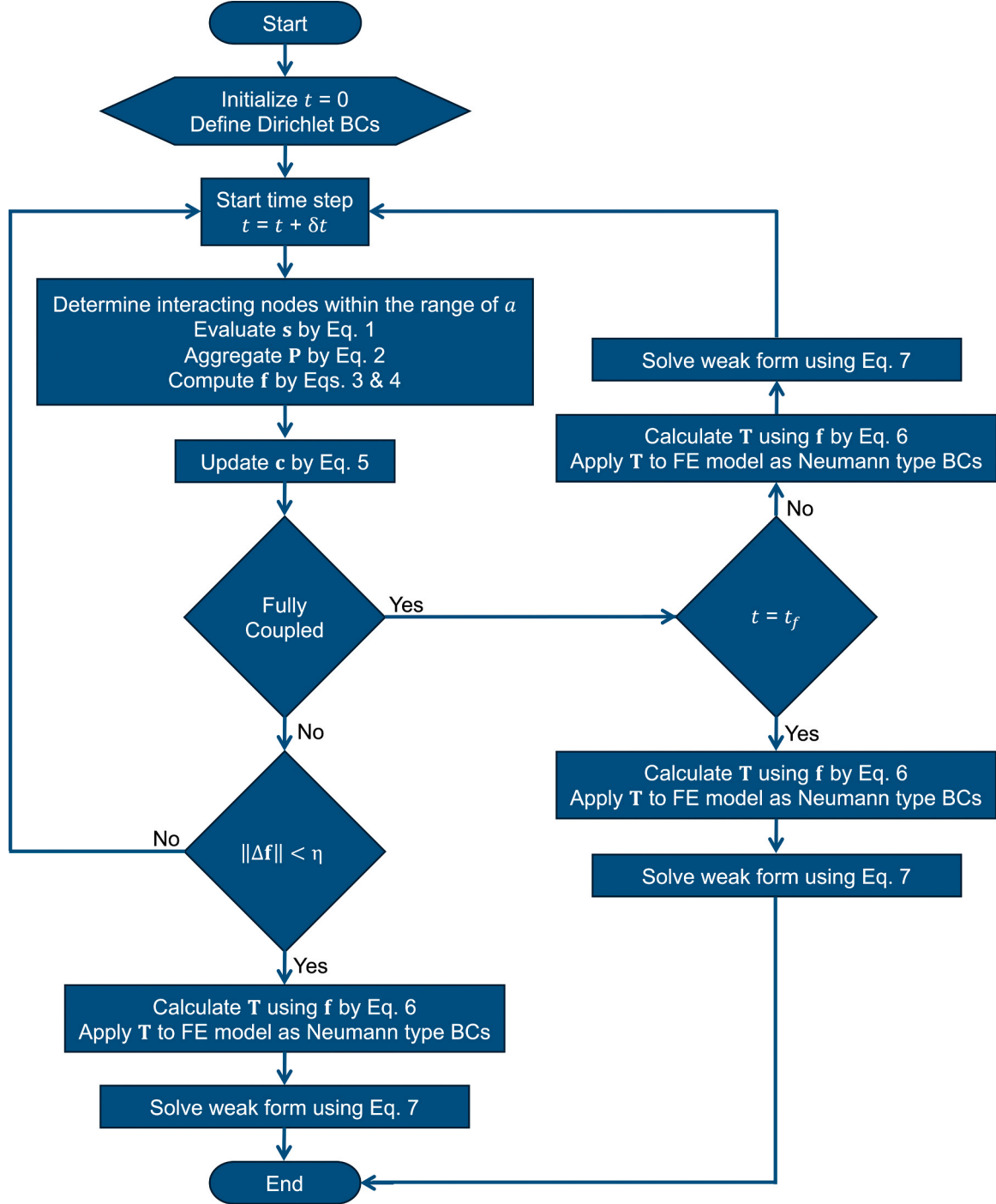


Fig. 2. Computational flowchart showing integration of the FuzzyContact formulation within the finite element (FE) framework. At each time step, contact forces are computed, converted into surface tractions, and imposed as Neumann boundary conditions to solve for displacement.

Newton's second law given in Equation 5. This acceleration-based update governs translational motion using the resultant external forces, $\mathbf{F}$, and contact forces, $\mathbf{f}$, to update the body's spatial coordinates over a pseudo-time increment, δt . The mass, M , and damping coefficient, ζ , provide numerical stabilization, ensuring convergence toward quasi-static equilibrium.

The algorithm advances through explicit numerical increments. At each increment, the contact search procedure identifies all interacting nodal pairs based on the proximity criterion, a , and the FuzzyContact law defined by Equations 1–3 computes the nodal contact forces, $\mathbf{f}$. The geometric configuration, $\mathbf{c}$, is then updated using the relaxation scheme in Equation 5. At the conclusion of this step, the key state variables are stored to provide the initial conditions for the next increment. These history variables are primarily the updated nodal contact forces, stored as $\mathbf{f}$, and the body's spatial coordinates, $\mathbf{c}$. This iterative process, where each step builds directly upon the converged state of the previous one, establishes the history-dependent nature of the formulation, ensuring a continuous and physically consistent evolution of the contact interface. Once the convergence criterion is met, the discrete nodal forces are converted into a continuous surface traction field, $\mathbf{T}$, via Equation 6 and imposed on the variational problem as Neumann-type boundary conditions, while essential Dirichlet constraints prescribe displacements at fixed regions. The weak form of the linear elasticity equations given in Eq. 7 is then solved to obtain the internal displacement and stress fields corresponding to the stabilized contact configuration.

As depicted in the flowchart, the integration of the weak-form analysis within the contact algorithm can follow either a fully coupled or a decoupled strategy. In the fully coupled approach, the weak form is solved within each contact iteration, continuously updating the internal stress and deformation fields as contact forces evolve until the termination time step, t_f , is reached, offering superior robustness for dynamic problems involving large deformations or highly nonlinear materials. In contrast, the decoupled mode, separates these tasks for computational efficiency where the algorithm first iterates exclusively on contact mechanics to achieve a stable force equilibrium. This iterative cycle continues until the force residual, $\|\Delta \mathbf{f}\|$, falls below the prescribed tolerance, η , indicating equilibrium of the contact system and only then performs a single weak-form solve to compute the final stress and strain distribution. This decoupled hybrid framework provides a numerically stable and efficient means to track evolving contact configurations with high fidelity, particularly suited to simulations involving stiff bodies and complex interfaces where contact search dominates computational cost.

2.3. Parameter selection and sensitivity

The physical realism and numerical stability of the FuzzyContact formulation are governed by a set of key parameters that control the behaviour of the sigmoidal traction law and the overall numerical response. These include the proximity range, a , transition parameter, b , and power exponent, m , which collectively define the shape and stiffness evolution of the traction-distance relation, as well as auxiliary parameters such as the damping coefficient, ζ , friction coefficient, μ , and penalty force scale, $\mathbf{k}$, that influence convergence and constraint enforcement. These parameters are calibrated through a structured process that balances physical fidelity and numerical robustness. The convergence threshold, η , on the other hand, is typically set to a value approaching zero to ensure that the contact system reaches an equilibrium state with negligible force imbalance, thereby achieving numerical precision consistent with exact static equilibrium. Scaling the sigmoidal parameters, a , b , and m , in a coordinated manner ensures a consistent physical response and maintain numerical stability.

The proximity range parameter, a , defines the proximity distance for contact interaction and is directly linked to the local mesh resolution of the contacting surfaces. Its primary role is to ensure computational efficiency by limiting the contact search to only geometrically proximate

nodes. If a is too small, smaller than a typical element size, nodes on opposing surfaces may fail to detect each other, preventing contact from being established. If a is excessively large, the computational cost of the contact search increases significantly, and there is a risk of generating spurious interactions between non-adjacent surfaces. Therefore, a is directly linked to the local mesh resolution, or characteristic element size, h .

The transition parameter, b , controls the sharpness of the force escalation once nodes are within the proximity range a . It defines the length scale over which the contact stiffness transitions from near-zero to its maximum gradient. The investigations on sensitivity determined that a value for b in the range of $0.05a$ to $0.5a$ provides an optimal balance. If b is too small, approaching zero, the sigmoidal function becomes a near-step function, reintroducing the numerical stiffness and potential for force oscillations that this formulation is designed to avoid. If b is too large, the contact response becomes overly soft. This balance is particularly crucial during mesh refinement. While a small b/a ratio is effective for coarse meshes, this ratio is strategically increased for finer meshes. This scaling prevents the transition zone from becoming infinitesimally small, which would reintroduce the instability of a step-function.

The power exponent, m , governs the steepness of the sigmoidal traction law, controlling how rapidly the repulsive force increases as the inter-nodal distance approaches the transition range. A low m value produces a gradual, compliant contact response (soft contact), whereas a high m value yields a sharper transition that more strongly penalizes interpenetration, approximating hard contact. This calibration is also linked to mesh resolution. If m is held constant as the mesh is refined, proximity range, a , decreases and the repulsive force would escalate over a drastically smaller absolute distance, introducing severe numerical stiffness and causing convergence instabilities. Therefore, scaling m proportionally with the proximity range, a , maintains a consistent physical response and ensure numerical stability and ensures the relative steepness of the force curve remains consistent with the element size.

The selection of the damping coefficient, ζ , is guided by the need for efficient convergence to the quasi-static equilibrium. As this formulation employs a dynamic relaxation scheme for the rigid-body motion, the ideal value of ζ provides near-critical damping to the system's translational modes. A value that is too low would permit non-physical oscillations, requiring many increments to settle, while a value that is too high (overdamped) would needlessly slow the convergence.

The selection of the penalty force scale, $\mathbf{k}$, is governed by a fundamental trade-off between physical accuracy and numerical stability. Its primary role is to enforce the non-penetration constraint by generating a sufficiently large repulsive force. If $\mathbf{k}$ is too low, the contact response becomes overly soft, permitting unphysical interpenetration. Conversely, an excessively high value can lead to numerical instability and ill-conditioning and convergence failure. In this formulation, the penalty force scale, $\mathbf{k}$, has units of force (N). A robust initial estimate for this parameter is a value proportional to $E \times h^2$, yielding the necessary units of force. This estimate is physically motivated as the product of the material's Young's modulus, E (in units of N/m²), and a characteristic element area, h^2 (in units of m²), yields a value with units of force. This characteristic force represents a physically meaningful starting point, scaling the penalty response with both the material's intrinsic stiffness and the mesh discretization. The optimal $\mathbf{k}$ for a given simulation is then finalized through a structured sensitivity analysis to ensure it is high enough to strictly enforce the constraint while maintaining robust numerical convergence.

The friction coefficient, μ , governs the ratio between tangential and normal resistance in the contact response and thus directly influences the onset and magnitude of stick-slip behaviour. In the present formulation, the friction coefficient is treated as a physical material parameter rather than a numerical tuning factor and can be specified either iso-

tropically or directionally. For metallic or ceramic interfaces where adhesive effects are negligible, typical values between 0.3 and 0.6 provide realistic frictional limits.

3. Benchmarks and model validation

The predictive capability and numerical robustness of the Fuzzy-Contact method were assessed through a structured series of five benchmarks designed to verify accuracy, evaluate stability, and validate physical fidelity. These benchmarks were arranged in increasing order of complexity, covering analytical, numerical, and experimental domains. The first benchmark is a patch test, which verifies the model's ability to reproduce the analytical Hertzian contact response and maintain stress continuity across the interface. The second benchmark investigates combined normal and tangential loading, evaluating the formulation's ability to capture smooth traction transitions and implicit stick-slip behaviour through its continuous, history-dependent friction formulation. The third benchmark involves a V-indenter problem with a sharp contact geometry, serving to test the method's robustness against local stress concentration and its agreement with a commercial finite element solver. The fourth benchmark provides experimental validation through comparison with in-situ neutron diffraction measurements performed on a compressed steel cylinder. The fifth benchmark examines a connected assembly configuration, used to assess mesh-convergence characteristics and the scalability of the formulation in multi-body contact scenarios. The computed elastic strain components are directly compared with measured values to confirm the physical accuracy and predictive fidelity of the formulation under realistic loading conditions. Collectively, these five benchmarks provide a comprehensive assessment of the Fuzzy-Contact method across analytical, numerical, and experimental levels, demonstrating its reliability, scalability, and versatility for generic contact simulations in solid mechanics.

The five benchmark problems were solved using a common set of parameters for the sigmoidal traction function where the proximity range, a , was linked to the characteristic element size, h . To create a smooth transition that effectively simulates hard contact, the parameters were scaled together. In these simulations, the transition parameter was kept constant to ensure computational efficiency, as smaller values of this parameter required significantly finer time steps and resulted in substantially higher computational cost. The values for the power exponent, m , proximity range, a , and transition parameter, b , were set as follows:

- When a was 0.5 mm, m was 110, and b was $0.05a$.
- When a was 0.25 mm, m was 55, and b was $0.1a$.
- When a was 0.125 mm, m was 27.5, and b was $0.2a$.
- When a was 0.0625 mm, m was 13.75, and b was $0.4a$.

A constant damping coefficient of 0.5 kg/s was applied in all cases. This value was determined empirically to provide robust and efficient convergence for the class of problems presented. While the optimal damping coefficient is system-dependent, this value demonstrated broad applicability across the benchmarks, effectively minimizing non-physical oscillations without overdamping the system's relaxation to a quasi-static equilibrium.

Although the general formulation given in Equation 3 allows for orthotropic definitions, isotropic models for the friction coefficient, μ , and penalty force scale, k , were employed in the benchmarks presented in this study to simplify the analysis. Specifically, the friction coefficient, μ , was set to 0.5 in Benchmark 2 to evaluate the formulation's ability to capture tangential stick-slip behaviour. Similarly, the penalty force scale was treated as isotropic by setting its directional components to a single value, with the magnitude of this value being specific to each problem. For detailed post-processing and visualization, the complete solution files for each benchmark, containing all components of nodal displacement, stress, and strain fields, are available in VTU format in the

designated data repository.

3.1. Benchmark 1: patch test in terms of Hertzian contact

The first benchmark evaluates the capability of the FuzzyContact method to reproduce classical Hertzian contact behaviour through a patch test involving elastic contact between a sphere with a radius of 1 mm and a rectangular block with dimensions of $4 \times 4 \times 2 \text{ mm}^3$. This configuration serves as a fundamental verification of the model's ability to capture the nonlinear load-displacement response and stress distribution predicted by analytical theory. Both bodies are treated as linear elastic, and the contact interaction is governed solely by the sigmoidal traction law without predefined surface definitions or contact pairs. The applied normal load is gradually increased to establish the equilibrium contact condition, while the resulting z -component of elastic deformation is monitored to quantify the displacement field.

The patch test serves as a fundamental benchmark in contact mechanics to verify the formulation's capability to reproduce uniform contact pressure distributions, ensure consistent load transfer, and satisfy mechanical equilibrium under idealized conditions. This assessment establishes the reliability of the proposed approach for more complex contact problems through direct comparison with analytical solutions. Both bodies were modelled as aluminium, with a density of 2700 kg/m^3 , Young's modulus of 70 GPa, and Poisson's ratio of 0.33, consistent with linear elastic assumptions. An external compressive load of -1000 N was applied along the z -axis to the spherical body. The contact interaction was solved using the quasi-static FuzzyContact formulation. The domain was discretized using tetrahedral elements with an average size of 0.25 mm, and an isotropic penalty force scale of 1000 N was determined. After quantifying the contact forces, top nodes of the spherical body and the bottom surface of the lower rectangular body were fully constrained in all translational directions to ensure global equilibrium during the subsequent finite element calculations. This setting ensured stable convergence and realistic contact pressure without excessive stiffness.

The simulation captured the complete elastic strain and stress tensors for both bodies, with displacement results presented in Fig. 4. The z -component of displacement is plotted on the undeformed geometries, showing a maximum downward displacement of $13.78 \text{ }\mu\text{m}$ in the lower rectangular body and a maximum upward elastic deformation of approximately $33.03 \text{ }\mu\text{m}$ in the spherical body. To facilitate a rigorous comparison with Hertzian theory, the validation analysis focused exclusively on the deformation of the spherical body. This approach was adopted because the lower rectangular body was modelled with finite dimensions, whereas the analytical solution assumes a semi-infinite half-space. The proximity of the fixed boundary in the finite model introduces structural stiffening that naturally reduces the displacement of the base compared to the infinite assumption. Modelling a domain sufficiently large to approximate a half-space would require excessive computational resources and is unnecessary for verifying the local contact behaviour.

Hertzian contact theory provides the analytical basis for validating the computed results, representing a classical solution for frictionless, elastic contact between non-conforming geometries. Developed by Heinrich Hertz [1], this theory predicts the pressure distribution, contact radius, and elastic deformation under the assumptions of linear elasticity, homogeneity, isotropy, small strains, and perfectly smooth surfaces. In the present benchmark, the contact between the spherical indenter and the flat block corresponds to the classical sphere-on-flat Hertzian case. For identical materials, the effective Young's modulus, E^* , is computed as 39277 N/mm^2 using its relationship with Young's modulus and Poisson's ratio, $E^* = E/(2(1 - \nu^2))$. Subsequently, the equivalent radius, R^* , is defined as 1.0 mm and the flat plate is treated as having an infinite radius of curvature. The contact radius, r_c , is determined to be 0.2673 mm when the total external load, L , is -1000 N

using Hertz contact radius equation, $r_c = (3LR^*/4E^*)^{1/3}$. According to the total elastic approach in Hertzian contact theory, $\delta = r_c^2/R^*$, indentation depth, δ , is obtained as 71.45 μm .

For two identical deformable bodies, this indentation is symmetrically distributed, giving an elastic approach of 35.7 μm for each body. In the numerical simulation, which explicitly models both the sphere and plate as deformable solids, the single body indentation, computed as the z-displacement of the sphere, is 33.03 μm . This value shows agreement with the analytical prediction for single-body indentation, deviating by just over 7.4 %. This confirms that the quasi-static FuzzyContact method accurately reproduces the analytical load-deformation response and the classical Hertzian behaviour for elastic, frictionless contact.

The computed contact pressure and deformation fields exhibit smooth, continuous transitions across the interface, free from numerical oscillations typically associated with discrete master-slave contact enforcement. The simulated contact radius and maximum indentation depth agree closely with the analytical Hertzian solution, confirming the accuracy of the proximity-based contact law. These results demonstrate that the proposed formulation faithfully reproduces classical elastic contact mechanics while maintaining numerical stability and geometric symmetry, thereby validating the core traction-distance relation underlying the FuzzyContact model.

3.2. Benchmark 2: combined normal and tangential loading

The second benchmark problem was designed to evaluate the FuzzyContact method's ability to capture the transition from normal to tangential contact, in decoupled mode, particularly in the presence of non-conforming meshes. The domain consists of a stationary body, $20 \times 10 \times 2 \text{ mm}^3$, and a moving body, $4.5 \times 4.5 \times 1 \text{ mm}^3$, which generates contact forces through prescribed motion. To test the robustness of the model, the bodies were meshed with tetrahedral elements, with average element size of 0.5 mm, as shown in Fig. 3b, featuring intentionally non-aligned nodes, a common feature of finite-element discretizations, as illustrated in Fig. 5a. Both bodies were defined as a stiff, isotropic linear elastic material, with a Young's modulus of 200 GPa and a Poisson's ratio of 0.3.

Normal contact was first established by applying a distributed compressive load of -2000 N along the z-axis to the upper body, achieving static equilibrium under pure normal pressure. Once equilibrium was reached, a tangential load of $+2000 \text{ N}$ was applied along the x-axis to initiate sliding, as shown in Fig. 5b. Although the explicit dynamic solver in FuzzyContact is capable of resolving transient and inertial effects, this benchmark focuses on validating the frictional contact behaviour under quasi-static conditions. To ensure quasi-static equilibrium, velocities were reset to zero at each increment, effectively suppressing cumulative acceleration and isolating the contact response from inertial artifacts. This procedure allows the contact algorithm to evolve based solely on geometric proximity and force history, consistent with the formulation's quasi-static assumptions. The top surface of the upper body and the bottom surface of the stationary lower body were fully constrained in all translational directions during the numerical finite element calculations after the quantification of contact forces, and, for computational efficiency, the finite element analyses were performed only at two representative stages of the contact interaction corresponding to the results presented in Fig. 6.

The tangential resistance was governed by normal contact pressure and the friction model where the contact surface normal was $\mathbf{N}(0, 0, 1)$. The simulation used an isotropic penalty force scale of 25 N providing a stable balance between constraint enforcement and convergence. The resulting tangential response reproduced the expected stick-slip transition predicted by Coulomb's law. During the initial loading phase, the tangential force increased linearly with displacement, corresponding to elastic sticking. Upon reaching the frictional limit, the tangential force saturated at approximately 1000.075 N, which agrees with the theoretical Coulomb limit of 1000.0 N within a negligible error that is less than 0.008 %. This precise agreement verifies that the contact algorithm correctly identifies the transition from static to kinetic friction. The force-displacement response, plotted in Fig. 5, provides a clear illustration of this process. The graph highlights the initial, linear region characteristic of the elastic sticking regime, which then seamlessly transitions to a constant force plateau as the contact enters the full sliding (slip) regime at the theoretical Coulomb limit.

Fig. 6 presents the distributions of displacement and stress

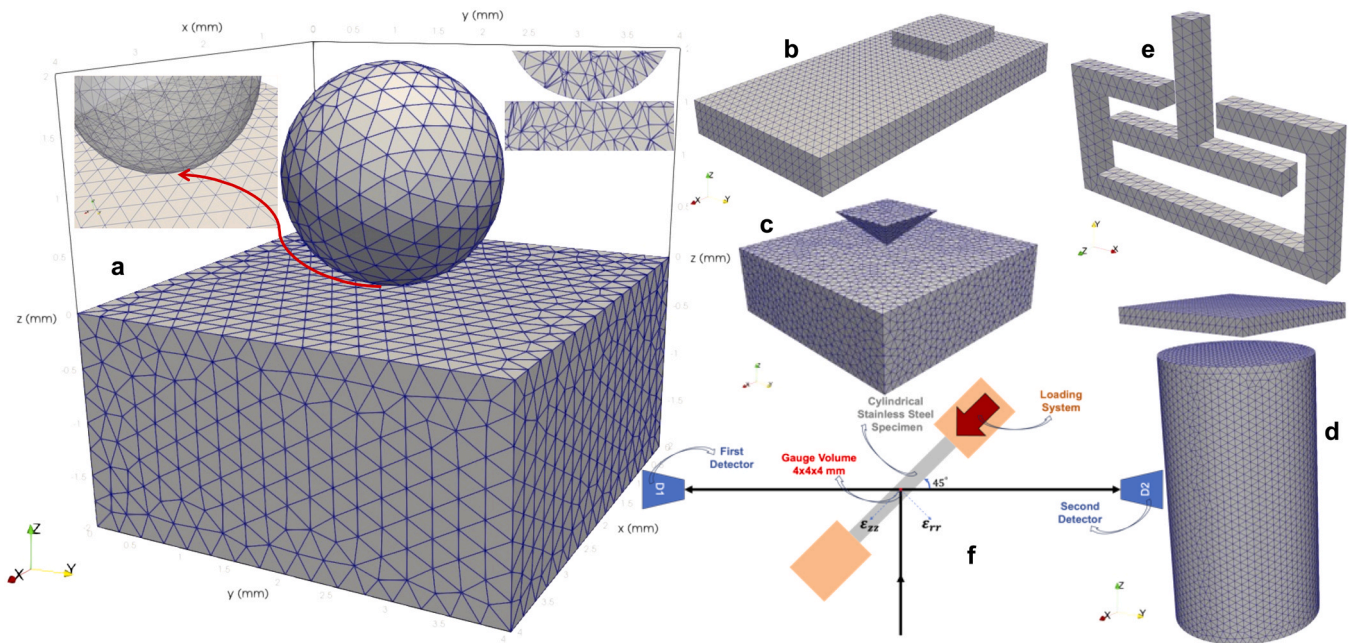


Fig. 3. Overview of the finite element meshes, and experimental configurations used for the benchmark problems. The domains shown correspond to (a) the patch test, (b) combined normal and tangential contact, (c) the V-indenter simulation, (d) the compressed cylinder for neutron diffraction validation, and (e) the connected assembly for mesh-convergence analysis. Sub-figure (f) provides a detailed schematic of the in-situ neutron diffraction setup, illustrating the loading configuration, incident beam, and detector geometry used for benchmark (d).

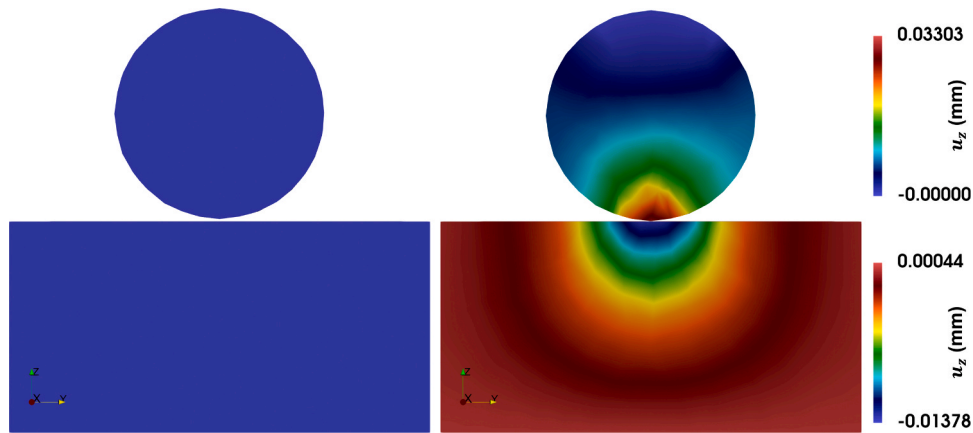


Fig. 4. Patch-test simulation results showing the z -component of elastic deformation under a -1000 N load. Downward deformation occurs in the lower rectangular body, while upward relative displacement in the spherical body reflects the reaction force at the contact interface.

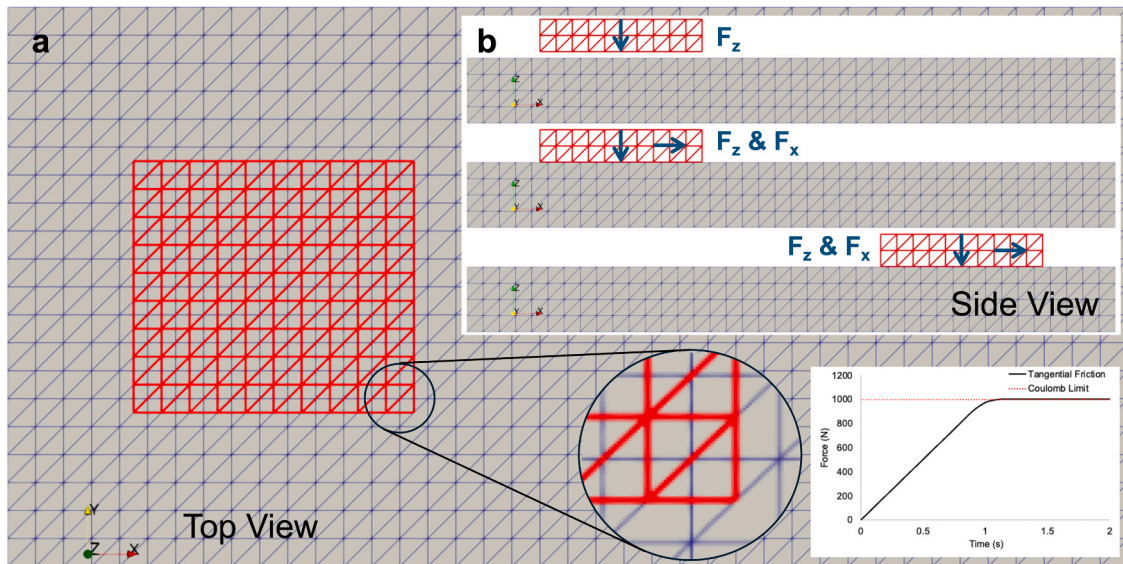


Fig. 5. Contact configuration for the combined normal-tangential loading benchmark. (a) Finite element meshes of the contacting bodies with intentionally non-matching surface nodes. (b) Schematic stages of the contact process that are initial state, normal compression, and subsequent tangential displacement inducing frictional effects.

components before and after tangential loading. At the onset of tangential motion, the x - and y -components of displacement were nearly symmetrical and localized around the contact edges. As sliding progressed, the x -component became dominant, the y -component weakened and spread laterally, and the z -component adjusted due to the evolving interface geometry. The computed stress fields remained continuous and physically consistent, showing balanced tensile and compressive zones across the interface. Localized perturbations in the displacement and traction fields reflect mesh-induced variations, which can be reduced by refining the surface discretization. Overall, the results confirm that the FuzzyContact method robustly reproduces coupled normal-tangential interactions, accurately capturing the elastic stick-slip behaviour even under non-conforming mesh conditions.

3.3. Benchmark 3: V-indenter problem and commercial solver validation

The third benchmark was designed to assess the FuzzyContact method's performance on non-flat, complex geometries with sharp features using a 3D V-indenter problem illustrated in Fig. 3c. To provide

a rigorous comparison, the results were cross-validated against the commercial finite element solver ABAQUS. The primary goal was to evaluate the formulation's ability to maintain symmetry and robustness when handling the localized contact and high-stress gradients that arise from such geometries.

The model consisted of a V-shaped indenter, formed from a 2×2 mm² square-based prism with a 1 mm height, pressed against a $6 \times 6 \times 2$ mm³ rectangular base plate. To isolate geometric effects from material asymmetry, both bodies were modelled as deformable, linear-elastic solids with identical properties typical aluminium: a Young's modulus of 70 GPa, a Poisson's ratio of 0.33, and a density of 2700 kg/m³. A compressive load of -90 N was applied along the z -axis until global force equilibrium was achieved, with the results shown in Fig. 7. To rigorously test algorithmic robustness, the contact interface was discretized with irregular tetrahedral elements, with average size of 0.25 mm, to intentionally introduce nodal asymmetry. The top surface of the V-indenter and the bottom surface of the base plate were fully constrained in all translational directions during the numerical finite element calculations after the quantification of contact forces. A stable

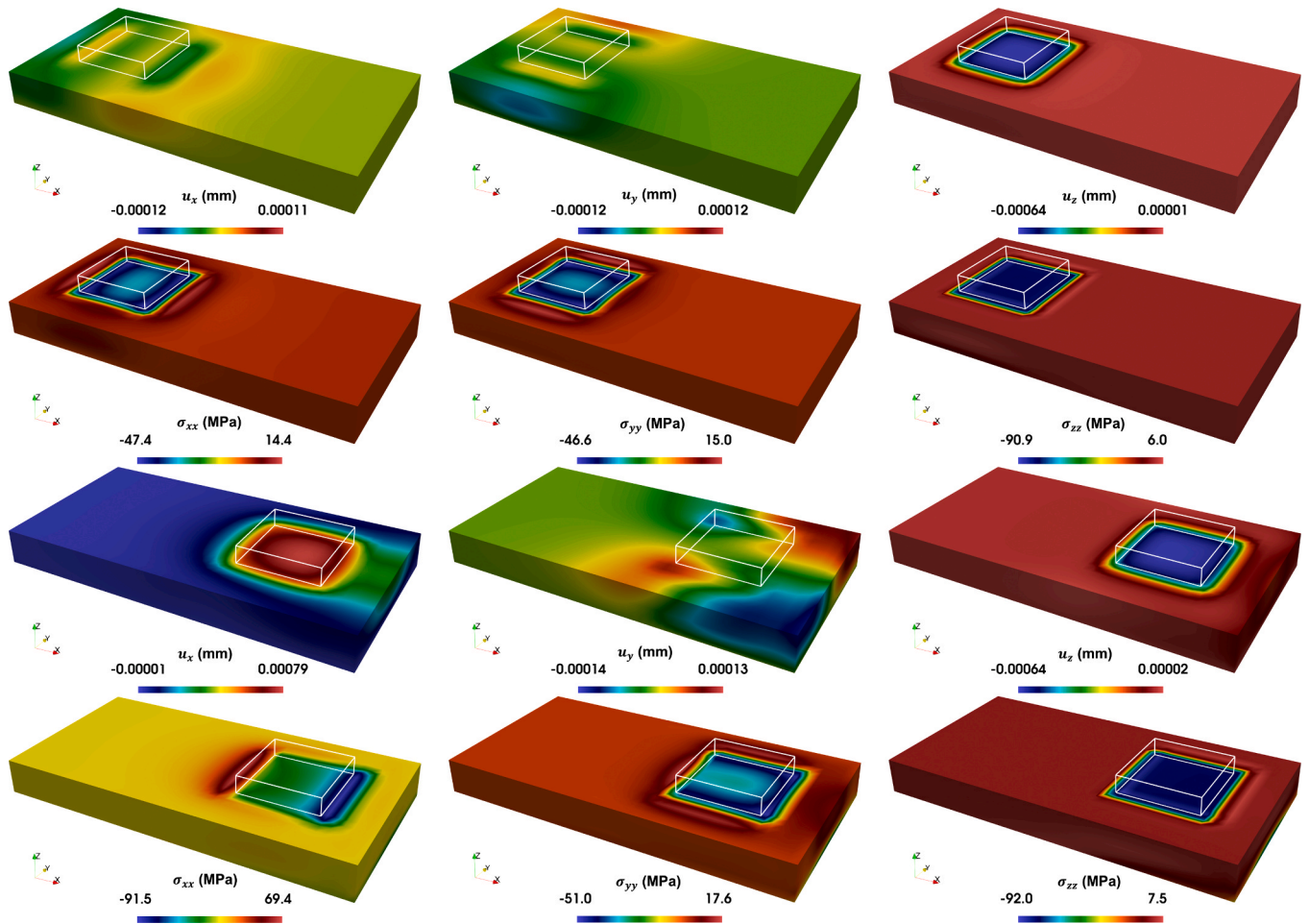


Fig. 6. Computed displacement and stress fields resulting from combined normal and tangential loading. Spatial distributions of the displacements, u_x , u_y and u_z , and the stress tensor components, σ_{xx} , σ_{yy} and σ_{zz} , due to contact forces after normal compression in the first two rows and after tangential motion in the last two rows.

and convergent solution was achieved using an isotropic penalty force scale of 250 N, which effectively minimized interpenetration.

The computed results revealed physically consistent stress distributions across the contact interface. Owing to the geometric stress concentration at the sharp apex, the maximum von Mises stress occurred directly beneath the indenter tip. Despite both bodies sharing identical material properties, the indenter's sharp geometry induced a region of high stress concentration in the underlying plate. Within the plate, the stress contours formed a fan-shaped pattern extending from the contact region, consistent with classical analytical descriptions of sharp indentation mechanics [72]. The symmetry of the stress field and the absence of numerical artifacts confirm that the formulation accurately handles contact for complex, non-planar geometries without requiring a master-slave definition.

To further validate the formulation, the V-indenter problem was also solved using the commercial finite element solver ABAQUS/Standard. For a direct and fair comparison, the ABAQUS model was built using the same mesh, consisting of C3D4 linear tetrahedral elements, and the interaction was defined using a surface-to-surface contact formulation with a standard penalty enforcement method. A direct comparison of the results required careful consideration of the distinct numerical architectures of the two solvers, as each performs optimally under different control modes for this class of problem. ABAQUS/Standard, as a monolithic implicit solver, can exhibit severe convergence difficulties for sharp contact problems when under direct force control. Consequently, the standard and robust practice of using displacement control

was employed to obtain a stable reference solution. In contrast, the hybrid dynamic relaxation scheme of the FuzzyContact method is inherently load-driven, making force application the most direct and natural mode of operation. This necessary difference in control modes does not compromise the validity of the comparison. Because the benchmark problem is linear-elastic and quasi-static, the final equilibrium state is path-independent; the stress and displacement fields depend only on the final converged configuration, not the manner in which the load was applied. Therefore, a meaningful and rigorous comparison was established by evaluating both frameworks at an equivalent mechanical state.

The proposed FuzzyContact method adopts a modified penalty approach that integrates geometric-proximity weighting with a sigmoidal traction law. The contact force magnitude is expressed as a continuous function of the inter-nodal distance, ensuring a smooth transition between the initial contact and full compression states. In contrast to the linear stiffness law used in ABAQUS, the sigmoidal representation yields a nonlinear, continuously differentiable stiffness curve, enhancing stability and robustness, particularly under irregular or sharp contact geometries. Both methods maintain computational efficiency by avoiding the Lagrange-multiplier enforcement of non-penetration constraints, thereby reducing system size and ensuring rapid convergence.

To create a direct basis for comparison, the ABAQUS simulation was performed using a displacement-controlled approach. A targeted downward displacement was applied to the indenter's top surface. This

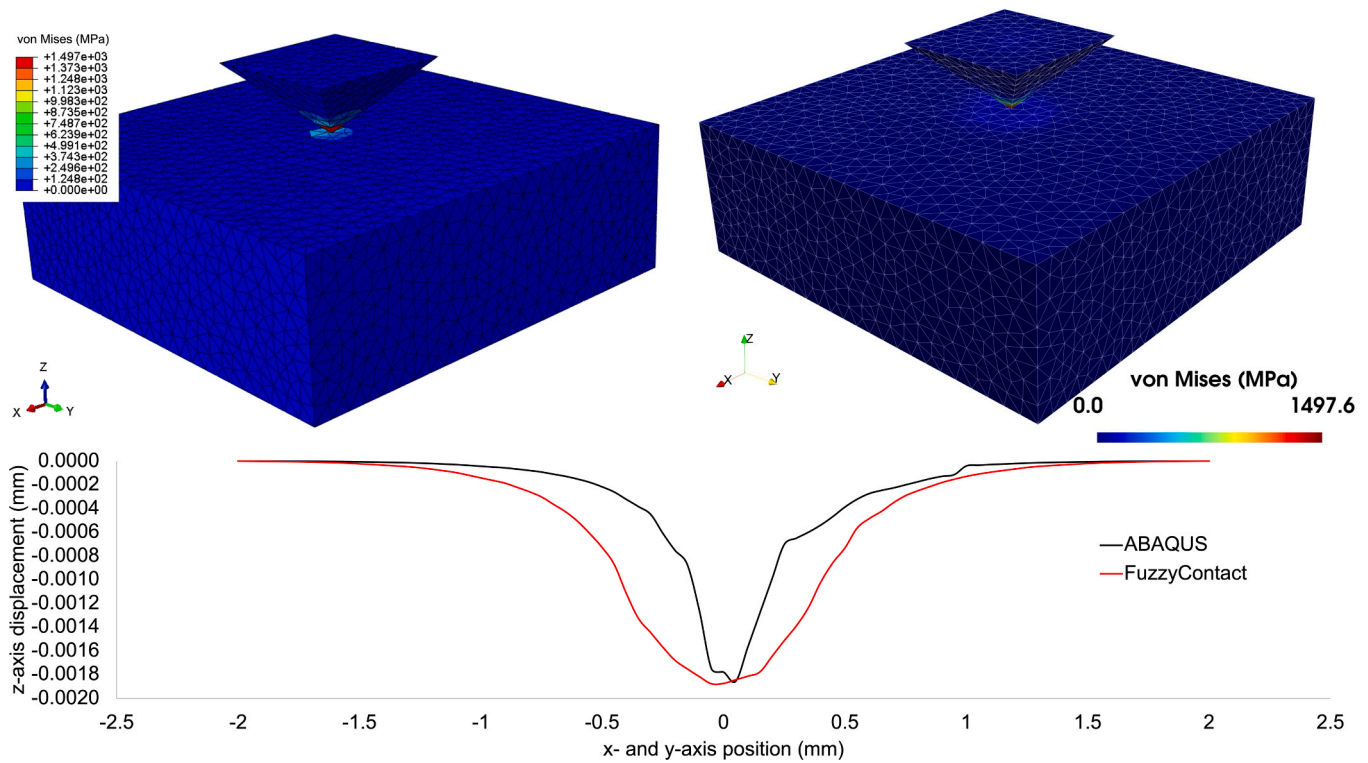


Fig. 7. Simulation results of the V-indenter problem, comparing von Mises stress distributions and displacement profiles from ABAQUS and the FuzzyContact method. The displacement is plotted along a diagonal path on the base plate's top surface, from coordinates (-2 mm, -2 mm) to (2 mm, 2 mm).

magnitude was rigorously calibrated to account for the initial inter-nodal gap and the compliance of the contact interface, ensuring that the resulting reaction force generated a maximum von Mises stress equivalent to that obtained from the load-controlled FuzzyContact simulation. This equivalence ensures that both models were evaluated at comparable peak stress states, facilitating a valid assessment of the deformation fields. The comparison between ABAQUS and FuzzyContact results shows agreement in the von Mises stress distributions obtained for the V-indenter configuration as shown in Fig. 7. Both models predict the highest stresses at the indenter tip, an expected outcome of geometric concentration under sharp contact loading. The simulations used identical meshes to ensure comparability, and the stress contours exhibited nearly identical magnitudes and spatial distributions. Although the computed peak stresses locally exceed the nominal yield strength of the material, this outcome is expected because the analysis is performed under a purely linear-elastic assumption. Consequently, the comparison between FuzzyContact and ABAQUS remains fully valid within the elastic regime. In elastic analyses of sharp contact geometries, the theoretical stress field is singular at the indenter apex. Therefore, stresses near such features can rise far above the actual material yield limit. The present comparison nonetheless remains meaningful, as both models are evaluated within the same linear-elastic framework, and the correlation of their stress fields confirms the fidelity of the proposed contact formulation under complex geometric conditions.

The direct comparison of the displacement fields given in Fig. 7 further validates the formulation's accuracy. In both the FuzzyContact and ABAQUS simulations, the maximum z -displacement on the base plate's surface reached a peak value of approximately 0.00185 mm. While the magnitudes are in agreement, a subtle but significant difference in the displacement profile highlights the distinct nature of the two contact algorithms. The displacement profile from the ABAQUS solution exhibits a sharper gradient at the contact boundary. This is a direct consequence of its more traditional penalty enforcement, where contact

pressure is activated once a discrete interpenetration threshold is met. In contrast, the FuzzyContact result shows a smoother, more continuous transition to the peak displacement. This behaviour is inherent to the sigmoidal traction law, which defines the contact force as a continuously differentiable function of inter-nodal proximity, eliminating the sharp onset of contact. This comparison not only confirms the agreement in the predicted deformation magnitude but also showcases the intended numerical character of the FuzzyContact method, which is specifically designed to provide such smooth and robust solutions for sharp contact problems.

3.4. Benchmark 4: experimental neutron diffraction validation

The fourth benchmark provides a direct experimental validation of the FuzzyContact method by comparing its predictions against in-situ neutron diffraction data. The physical experiment involved compressive loading of an 18 mm-high, 10 mm-diameter stainless-steel cylinder, with internal elastic strains measured within a $4 \times 4 \times 4$ mm³ gauge volume using neutron diffraction technique using the experimental setup at the Engin-X beam line of the ISIS Neutron and Muon Source is shown schematically in Fig. 3f. The corresponding numerical simulation of FuzzyContact replicated these conditions, using the meshed domain presented in Fig. 3d. This benchmark is critical as it establishes a direct correspondence between computational predictions and physically measured strain fields, underscoring the importance of validating numerical models against high-fidelity experimental data to ensure predictive reliability. The results of this comparison are presented in Fig. 8.

In the experiment, a stainless-steel cylindrical specimen was subjected to controlled compressive loading while internal elastic strains were measured using neutron diffraction. Axial and radial strain components were recorded as functions of the applied stress, obtained by dividing the applied normal force by the specimen's cross-sectional area. Within the linear-elastic regime, the strain-stress relationships were fitted using linear regression, yielding the effective elastic constants of

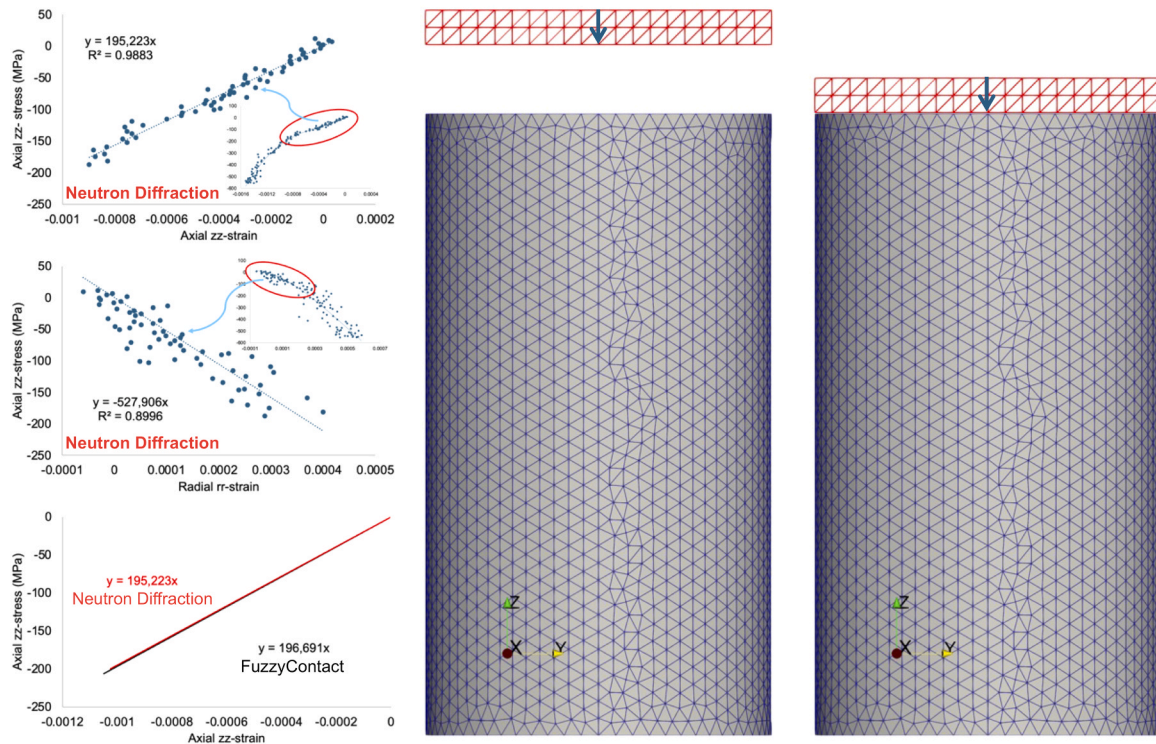


Fig. 8. Direct comparison of experimental and simulated stress-strain behaviour from the in-situ neutron diffraction benchmark. The plots show agreement between the neutron diffraction measurements and the FuzzyContact predictions as given with solid line for both axial and radial components.

the material. From the slope of the axial strain-stress curve, as illustrated in Fig. 8, an effective Young's modulus of 195.223 GPa was obtained, while the ratio of radial to axial strain slopes provided an effective Poisson's ratio of 0.3698. These experimentally derived properties were subsequently used as input parameters in the contact simulation to ensure consistency between physical and numerical models.

The corresponding numerical simulation reproduced the experimental setup by modelling the contact between the cylindrical specimen and a rectangular loading platen with dimensions $10 \times 10 \times 2 \text{ mm}^3$. The plate represented the rigid loading surface from the experiment and was subjected to the distributed compressive forces measured during testing. The contact problem was solved using the quasi-static Fuzzy-Contact formulation, employing an isotropic penalty force scale of 250 N which provided stable convergence and accurate enforcement of non-penetration without over-stiffening. The model domains were discretized with tetrahedral elements of average size 0.5 mm, ensuring sufficient spatial resolution near the contact interface.

The boundary conditions in the simulation were defined to replicate the physical experiment. The bottom face of the cylinder was fully constrained in all translational directions to prevent rigid-body motion, while the top surface was subjected to the distributed compressive load measured during the test. The lateral curved surface was left traction-free, allowing unrestricted radial expansion consistent with the Poisson effect observed experimentally. This configuration ensured that the boundary conditions accurately represented the unconfined uniaxial compression conditions in the neutron diffraction experiment.

Eight separate simulations were performed at distributed load states ranging from 2825 N to 22600 N with a step size of 2825 N to match the experimental force increments. The resulting internal stress distributions were evaluated within the same central gauge volume investigated by neutron diffraction. This allowed for a direct comparison between the axial stress-strain responses from the simulation and the experiment. The numerical results closely matched the experimental trends within the linear-elastic region as illustrated in Fig. 8. From the simulation's output, an effective Young's Modulus of 196.691 GPa was derived,

confirming that the proposed formulation accurately reproduces the elastic deformation and load transfer mechanisms observed experimentally. This benchmark primarily validates the formulation's accuracy for normal contact conditions. The close agreement between simulation and experiment confirms the formulation's capability to predict realistic elastic response under compressive contact loading.

3.5. Benchmark 5: contact in a connected assembly and mesh convergence

The fifth benchmark analysed contact forces within a connected multi-body assembly, shown in Fig. 3e, to investigate contact interactions in a geometrically complex configuration. The study of contact mechanics in interconnected bodies is of paramount importance in numerous engineering applications, as it governs the load transfer, stress distribution, and overall structural integrity. The problem consists of two distinct bodies: an inner, inverted T-shaped component and an outer enclosure surrounding the T-section's horizontal arm, effectively preventing its separation. Both the inner T-shaped and outer enclosing bodies possess cross-sections of $1 \times 1 \text{ mm}^2$ and are positioned within an overall simulation domain measuring $13 \times 9 \times 1 \text{ mm}^3$. In the simulation, the inner T-shaped body was subjected to a prescribed motion along the positive y-axis at a constant velocity of 0.001 mm/s until contact forces reduced its velocity to zero, achieving a stable equilibrium state. After quantifying the contact forces, the top surface of the inner body and the bottom surface of the outer body were fully constrained in all translational directions to ensure global equilibrium during the subsequent finite element calculations.

The connected assembly benchmark was solved using four progressively refined mesh configurations with average element sizes of 0.5 mm (coarse), 0.25 mm, 0.125 mm, and 0.0625 mm (fine) to formally assess spatial convergence. For each mesh, the isotropic penalty force scale was scaled consistently with mesh refinement, determined through a preliminary sensitivity analysis, as 0.32 N, 0.129 N, 0.04063 N, and 0.01223 N, respectively, to ensure consistent contact constraint

enforcement and numerical stability. In all cases, the proposed model successfully captured the contact interactions and reached a stable equilibrium state.

Notably, all four simulations converged to the same total contact force of 11.174 N. This perfect agreement is not a numerical coincidence but a direct consequence of the benchmark's design as a global equilibrium problem. The total contact force represents the global reaction necessary to balance the internal elastic stresses produced by wedging the inner body into the fully constrained outer body. Because this global reaction is dictated by the overall stiffness of the assembly, it remains largely independent of the local contact discretization. This outcome therefore validates the model's ability to consistently enforce global equilibrium across different mesh densities. While the total force remained constant, the local von Mises stress distributions exhibit

visible refinement-dependent differences, as shown in Fig. 9, reflecting the improved resolution of stress concentrations and geometric details at finer mesh scales.

A quantitative analysis of the mesh-convergence study was performed by extracting the maximum von Mises stress and displacement magnitude from the results of each mesh resolution, which are plotted in Fig. 9. Both key metrics approach stable, asymptotic values as the element size decreases from 0.50 mm to 0.0625 mm, demonstrating the systematic convergence of the solution. Specifically, the relative changes in both maximum stress and displacement fall below 5.0 % when the two finest mesh resolutions of 0.125 mm and 0.0625 mm, indicating that the solution has reached mesh-independent behaviour. As the mesh is refined, localized stress concentrations at the contact regions become more accurately resolved, confirming the convergence and mesh-

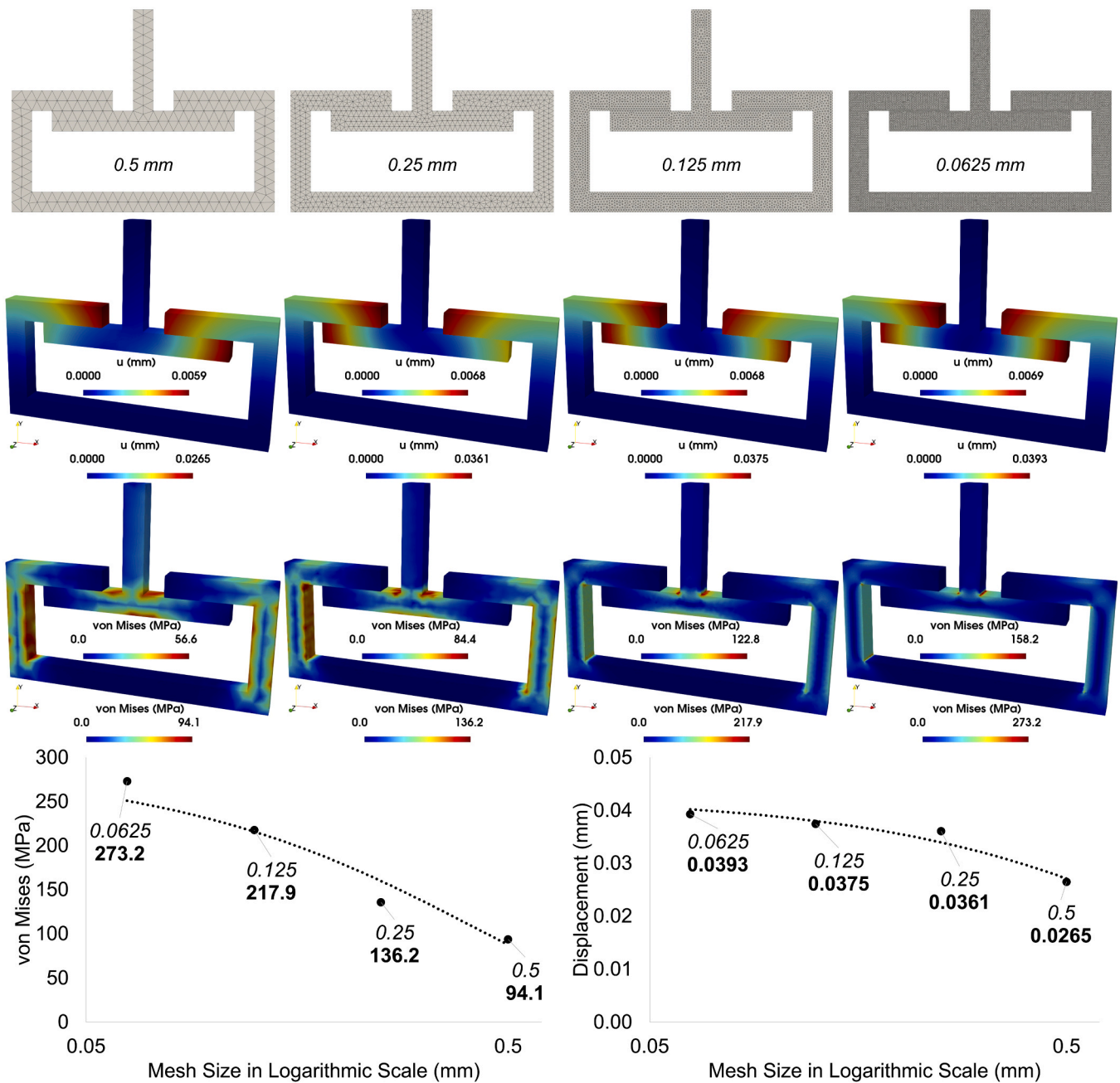


Fig. 9. Mesh-convergence analysis for the connected-assembly benchmark was performed using four mesh densities with average element sizes of 0.5 mm, 0.25 mm, 0.125 mm, and 0.0625 mm. The top row shows the corresponding finite element mesh configurations, the middle rows present the displacement magnitude, u , and von Mises stress fields, and the bottom row displays the distribution of the highest displacement and stress values as functions of element size on a logarithmic scale.

independent behaviour of the proposed formulation.

3.6. Assessment of predictive accuracy

A comprehensive quantitative validation was performed to assess the predictive accuracy of the FuzzyContact method against analytical, numerical, and experimental reference models. The benchmark problems that provide cross-validation are the patch test, tangential frictional contact, V-indenter indentation, and the experimental validation based on neutron diffraction data. The results are summarised in Table 1, which presents the reference values, FuzzyContact predictions, and the corresponding relative differences.

In the patch test, FuzzyContact achieved a total indentation depth of 33.03 μm , compared to 35.7 μm predicted by Hertzian theory, yielding a relative deviation of 7.48 %. This agreement confirms that the formulation reproduces the classical normal contact response with high fidelity and properly captures the curvature-induced deformation under elastic compression.

The tangential contact benchmark further validated the implementation of the frictional law within the formulation. The predicted maximum frictional force was 1000.075 N, in virtually perfect agreement with the 1000 N expected from the Coulomb law, corresponding to an error of only 0.0075 %. This demonstrates that the FuzzyContact framework preserves the correct frictional limit and transitions smoothly between stick and slip regimes.

The V-indenter benchmark provided a more complex 3D contact scenario for assessing the formulation's ability to capture localised deformation fields. The indentation depth predicted by FuzzyContact was 1.876 μm , compared to 1.856 μm obtained from ABAQUS, corresponding to a 1.077 % difference, indicating excellent consistency in the deformation magnitude. This not only verifies the quantitative accuracy of the formulation but also highlights its numerical robustness in treating sharp and complex contact geometries.

Finally, the experimental benchmark, based on the effective Young's modulus measured by neutron diffraction, yielded an FuzzyContact prediction of 196.691 GPa, compared to the reference value of 195.223 GPa, resulting in a relative deviation of 0.75 %. This close agreement reinforces the physical reliability of the model when applied to real-world material systems and experimental data.

Overall, the results presented in Table 1 demonstrate that FuzzyContact achieves close agreement across multiple contact scenarios. The formulation consistently reproduces analytical, numerical, and experimental results accurately, confirming its robustness, accuracy, and general applicability to both idealised and complex contact mechanics problems.

4. Conclusions

This study presented FuzzyContact, a computationally efficient and robust formulation for simulating complex contact mechanics. By defining contact forces as a continuous function of inter-nodal distance, FuzzyContact eliminates the need for predefined master-slave surfaces and explicit contact pair identification, a long-standing challenge in conventional algorithms. A central innovation is its ability to capture the stick-slip frictional response within a unified penalty framework. The sigmoidal traction law, combined with history-dependent proximity updates, enables a smooth, differentiable transition from sticking to sliding, thereby enhancing numerical robustness and ensuring symmetric interaction between contacting bodies.

The formulation's fidelity and robustness were rigorously verified through a comprehensive campaign of analytical, numerical, and experimental benchmarks. The method accurately reproduced the classical Hertzian contact solution, captured tangential friction consistent with Coulomb's law, and matched the stress and displacement fields predicted by the commercial solver ABAQUS. Most notably, predictions of elastic strain in a compressed steel cylinder showed excellent

Table 1

Quantitative summary of benchmark results, comparing FuzzyContact predictions against reference values.

Benchmark Problem	Parameter Compared	Reference Value	FuzzyContact Result	Relative Difference
Patch Test	Single Body Indentation Depth (μm)	35.7 (Hertzian Theory)	33.03	7.48 %
Tangential	Max. Frictional Force (N)	1000.0 (Coulomb Law)	1000.075	0.0075 %
V-indenter	Indentation Depth (μm)	1.856 (ABAQUS)	1.876	1.077 %
Experimental	Effective Young's Modulus (GPa)	195.223 (Neutron Diffraction)	196.691	0.75 %

agreement with in-situ neutron diffraction measurements, confirming the model's predictive reliability in a real-world scenario. Across all validation cases, excluding the analytical solution, FuzzyContact reproduced reference results within approximately 1 % deviation, demonstrating its high accuracy with respect to numerical and experimental cases.

In demonstrating its accuracy, scalability, and general applicability, FuzzyContact provides a solid foundation for high-fidelity tribological simulations. Future work will aim to enhance physical realism by incorporating elastic-plastic material behaviour and rate-dependent friction laws. From a computational standpoint, the proximity-based search algorithm is highly amenable to parallelization, and its GPU implementation is expected to significantly accelerate force calculations. These advancements will pave the way for extending FuzzyContact to multiphysics tribological problems involving wear, adhesion, and thermo-mechanical coupling.

Statement of originality

The authors hereby declare that the submitted manuscript, entitled "FuzzyContact: A generic contact mechanics formulation via a modified penalty method governed by sigmoidal traction law," represents original research work that has not been previously published in any form (including conference proceedings, journal articles, or book chapters), either in print or electronic format, and is not currently under consideration for publication elsewhere. All data presented within the manuscript were generated by the authors. The research presented in this manuscript adheres to all applicable ethical guidelines and standards for scientific integrity. The authors affirm that they have fully disclosed any potential conflicts of interest. We understand that the journal may utilize plagiarism detection software to assess the originality of this submission.

CRedit authorship contribution statement

Fatih Uzun: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Alexander M. Korsunsky:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors acknowledge EuroHPC project grant EHPC-DEV-2024D04-047 for allowing the simulations to be performed on the

Luxembourg national supercomputer MeluXina and are also grateful to LuxProvide teams for their expert support. The authors also thank ISIS Neutron and Muon Source (Didcot, U.K.) for beamline allocation RB2410153 to ENGIN-X Beamline, and beamline scientist Tung Lik Lee and Cyril Besnard from The University of Oxford for their support with the neutron diffraction experiment.

Data availability

The experimental data supporting the findings of this study will be made openly available via the ISIS Neutron and Muon Source data portal following an embargo period ending in 2027 at <https://doi.org/10.5286/ISIS.E.RB2410153-1>. The numerical solution files, including displacement, strain, and stress fields generated using the proposed contact formulation, are available in the Mendeley Data repository at <https://doi.org/10.17632/tdgkkyr2f.2>.

References

- [1] Hertz H. On the contact of elastic solids. In: Lenard P, editor. New York: Macmillan and Co Limited; 1881. p. 146–63.
- [2] Johnson KL, Kendall K, Roberts AD. Surface energy and the contact of elastic solids. *Proc R Soc Lond A Math Phys Sci* 1971;324:301–13. <https://doi.org/10.1098/rspa.1971.0141>.
- [3] Muller VM, Derjaguin BV, Toporov YP. On two methods of calculation of the force of sticking of an elastic sphere to a rigid plane. *Colloids Surf* 1983;7:251–9. [https://doi.org/10.1016/0166-6622\(83\)80051-1](https://doi.org/10.1016/0166-6622(83)80051-1).
- [4] Derjaguin BV, Muller VM, Toporov YP. Effect of contact deformations on the adhesion of particles. *J Colloid Interface Sci* 1975;53:314–26. [https://doi.org/10.1016/0021-9797\(75\)90018-1](https://doi.org/10.1016/0021-9797(75)90018-1).
- [5] Greenwood JA, Williamson JBP. Contact of nominally flat surfaces. *Proc R Soc Lond A Math Phys Sci* 1966;295:300–19. <https://doi.org/10.1098/rspa.1966.0242>.
- [6] Rafei M, Ghoreishy MHR, Naderi G. Computer simulation of tire rolling resistance using finite element method: effect of linear and nonlinear viscoelastic models. *Proc Inst Mech Eng Part D J Automob Eng* 2019;233:2746–60. <https://doi.org/10.1177/0954407018804117>.
- [7] Wang W, Yan S, Zhao Y. Numerical and experimental studies of a radial truck tire with tread pattern. *Simulation* 2015;91:970–9. <https://doi.org/10.1177/0037549715608434>.
- [8] Mashadi B, Ebrahimi-Nejad S, Abbaspour M. A rolling resistance estimate using nonlinear finite element numerical analysis of a full three-dimensional tyre model. *Proc Inst Mech Eng Part D J Automob Eng* 2019;233:147–60. <https://doi.org/10.1177/0954407018802733>.
- [9] Patil SS, Karuppanan S, Atanasovska I, Wahab AA. Contact stress analysis of helical gear pairs, including frictional coefficients. *Int J Mech Sci* 2014;85:205–11. <https://doi.org/10.1016/j.ijmecsci.2014.05.013>.
- [10] Zhang ZX, Fong ZH, Li YH, Fang HS. A study of the contact stress analysis of cylindrical gears using the hybrid finite element method. *Proc Inst Mech Eng C J Mech Eng Sci* 2013;227:3–18. <https://doi.org/10.1177/0954406212444383>.
- [11] Lin T, Ou H, Li R. A finite element method for 3D static and dynamic contact/impact analysis of gear drives. *Comput Methods Appl Mech Eng* 2007;196:1716–28. <https://doi.org/10.1016/j.cma.2006.09.014>.
- [12] Wu Y, Wang J, Han Q. Contact finite element method for dynamic meshing characteristics analysis of continuous engaged gear drives. *J Mech Sci Technol* 2012;26:1671–85. <https://doi.org/10.1007/s12206-012-0416-5>.
- [13] Qin WJ, Guan CY. An investigation of contact stresses and crack initiation in spur gears based on finite element dynamics analysis. *Int J Mech Sci* 2014;83:96–103. <https://doi.org/10.1016/j.ijmecsci.2014.03.035>.
- [14] Zhao X, Fan W, Wang Z, Wen Z, Wang P. An explicit finite element approach for simulations of transient meshing contact of gear pairs and the resulting wear. *Wear* 2023;523:204802. <https://doi.org/10.1016/j.wear.2023.204802>.
- [15] Lißner M, Erice B, Alabort E, Thomson D, Cui H, Kaboglu C, et al. Multi-material adhesively bonded structures: characterisation and modelling of their rate-dependent performance. *Compos B Eng* 2020;195:108077. <https://doi.org/10.1016/j.compositesb.2020.108077>.
- [16] Alessandri C, Mallardo V. Crack identification in two-dimensional unilateral contact mechanics with the boundary element method. *Comput Mech* 1999;24:100–9. <https://doi.org/10.1007/s004660050442>.
- [17] Xu Y, Jackson RL. Boundary element method (BEM) applied to the rough surface contact vs. BEM in computational mechanics. *Friction* 2019;7:359–71. <https://doi.org/10.1007/s40544-018-0229-3>.
- [18] Hajesfandiari AR, Hajesfandiari A, Dargush GF. Size-dependent contact mechanics via boundary element analysis. *Eng Anal Bound Elem* 2022;136:213–31. <https://doi.org/10.1016/j.enganabound.2021.12.016>.
- [19] Keum B, Liu Y. Analysis of 3-D frictional contact mechanics problems by a boundary element method. *Tsinghua Sci Technol* 2005;10:16–29. [https://doi.org/10.1016/S1007-0214\(05\)70004-3](https://doi.org/10.1016/S1007-0214(05)70004-3).
- [20] Putignano C, Carbone G. A review of boundary elements methodologies for elastic and viscoelastic rough contact mechanics. *Phys Mesomech* 2014;17:321–33. <https://doi.org/10.1134/S1029959914040092>.
- [21] Rojek J. Contact modeling in the discrete element method. CISM International Centre for Mechanical Sciences, Courses and Lectures, 585. Cham: Springer; 2018. p. 177–228. https://doi.org/10.1007/978-3-319-90155-8_4.
- [22] Fleissner F, Gaugele T, Eberhard P. Applications of the discrete element method in mechanical engineering. *Multibody Syst Dyn* 2007;18:81–94. <https://doi.org/10.1007/s11044-007-9066-2>.
- [23] Li M, Venäläinen MS, Chandra SS, Patel R, Fripp J, Engstrom C, et al. Discrete element and finite element methods provide similar estimations for hip joint contact mechanics during walking gait. *J Biomech* 2021;115:110163. <https://doi.org/10.1016/j.jbiomech.2020.110163>.
- [24] Radjai F, Richefeu V. Contact dynamics as a nonsmooth discrete element method. *Mech Mater* 2009;41:715–28. <https://doi.org/10.1016/j.mechmat.2009.01.028>.
- [25] Yang C, Tartaglino U, Persson BNJ. A multiscale molecular dynamics approach to contact mechanics. *Eur Phys J E* 2006;19:47–58. <https://doi.org/10.1140/epje/e2006-00004-9>.
- [26] Hamad F, Giridharan S, Moormann C. A Penalty Function Method for Modelling Frictional Contact in MPM. *Procedia Eng* 2017;175:116–23. <https://doi.org/10.1016/j.proeng.2017.01.038>.
- [27] Gu RJ, Murty P, Zheng Q. Use of penalty variable in finite element analysis of contacting objects. *Comput Struct* 2002;80:2449–59. [https://doi.org/10.1016/S0045-7949\(02\)00330-9](https://doi.org/10.1016/S0045-7949(02)00330-9).
- [28] Zang M, Gao W, Lei Z. A contact algorithm for 3D discrete and finite element contact problems based on penalty function method. *Comput Mech* 2011;48:541–50. <https://doi.org/10.1007/s00466-011-0606-5>.
- [29] Telliskivi T, Olofsson U. Contact mechanics analysis of measured wheel-rail profiles using the finite element method. *Proc Inst Mech Eng F J Rail Rapid Transit* 2001;215:65–72. <https://doi.org/10.1243/0954409011531404>.
- [30] Lone AS, Harmain GA, Jameel A. Modeling of contact interfaces by penalty based enriched finite element method. *Mech Adv Mater Struct* 2023;30:1485–503. <https://doi.org/10.1080/15376494.2022.2034075>.
- [31] Perić D, Owen DRJ. Computational model for 3-D contact problems with friction based on the penalty method. *Int J Numer Methods Eng* 1992;35:1289–309. <https://doi.org/10.1002/nme.1620350609>.
- [32] Quyen VTB, Tien DN, Dung NN. A modified penalty function method for treating multi freedom constraints in finite element analysis of frames. *J Phys Conf Ser* 2020;1425:012097. <https://doi.org/10.1088/1742-6596/1425/1/012097>.
- [33] Wang D-yong, Chen X, Jiang S bin, Qi J lin, Peng L-yun. Elastoplastic analysis of solid structures using penalty-based couple stress finite element method within framework of Cosserat continuum. *J Cent South Univ* 2022;29:1320–31. <https://doi.org/10.1007/s11771-022-4997-6>.
- [34] Guo J, Liu D. Penalty finite element method for the 2D/3D unsteady 2 incompressible thermomicroscopic fluid equations. *Adv Appl Math Mech* 2024;16:209–34. <https://doi.org/10.4208/aamm.OA-2023-0238>.
- [35] Quyen VTB, Tien DN, Dung NN, Yaghoubi S, Shirani E, Pishevar AR, et al. Penalty function method for imposing nonlinear multi freedom and multi node constraints in finite element analysis of frame systems. *IOP Conf Ser Mater Sci Eng* 2020;869:052063. <https://doi.org/10.1088/1757-899X/869/5/052063>.
- [36] Quyen VTB, Tien DN, Huong NTL. Treatment of nonlinear multi freedom boundary constraints in finite element analysis of frame system using lagrange multiplier method. *IOP Conf Ser Mater Sci Eng* 2020;960. <https://doi.org/10.1088/1757-899X/960/2/022076>.
- [37] Sewerin F, Papadopoulos P. On the finite element solution of frictionless contact problems using an exact penalty approach. *Comput Methods Appl Mech Eng* 2020;368:113108. <https://doi.org/10.1016/j.cma.2020.113108>.
- [38] Ghosh G, Duddu R, Annavarapu C. On the robustness of the stabilized finite element method for delamination analysis of composites using cohesive elements. *Int J Comput Methods Eng Sci Mech* 2021;22:538–58. <https://doi.org/10.1080/15502287.2021.1896607>.
- [39] Bich QVT, Ngoc TD, Thuy VTT. Penalty function method for geometrically nonlinear buckling analysis of imperfect truss with multi-freedom constraints based on mixed FEM (EDP Sciences) E3S Web Conf 2023;410. <https://doi.org/10.1051/e3sconf/202341003028>.
- [40] Mota A, Koliesnikova D, Tezaur I, Hoy J. Fundamentally new coupled approach to contact mechanics via the Dirichlet–Neumann Schwarz alternating method. *Int J Numer Methods Eng* 2023;124:226–52. <https://doi.org/10.1002/nme.70039>.
- [41] Chu X, Chen T, Huang N-jing, Xiao Y-bin. Penalty method for a class of differential nonlinear system arising in contact mechanics. *Fixed Point Theory Algorithms Sci Eng* 2022;2022. <https://doi.org/10.1186/s13663-022-00727-6>.
- [42] Zavarise G, Wriggers P, Stein E, Schrefler BA. A numerical model for thermomechanical contact based on microscopic interface laws. *Mech Res Commun* 1992;19:173–82. [https://doi.org/10.1016/0093-6413\(92\)90062-F](https://doi.org/10.1016/0093-6413(92)90062-F).
- [43] Zavarise G, Wriggers P, Stein E, Schrefler BA. Real contact mechanisms and finite element formulation—a coupled thermomechanical approach. *Int J Numer Methods Eng* 1992;35:767–85. <https://doi.org/10.1002/nme.1620350409>.
- [44] Paggi M, Barber JR. Contact conductance of rough surfaces composed of modified RMD patches. *Int J Heat Mass Transf* 2011;54:4664–72. <https://doi.org/10.1016/j.ijheatmasstransfer.2011.06.011>.
- [45] Bonari J, Marulli MR, Hagmeyer N, Mayr M, Popp A, Paggi M. A multi-scale FEM-BEM formulation for contact mechanics between rough surfaces. *Comput Mech* 2020;65:731–49. <https://doi.org/10.1007/S00466-019-01791-3>.
- [46] Paggi M, Carpinteri A, Zavarise G. A unified interface constitutive law for the study of fracture and contact problems in heterogeneous materials. *Lecture Notes in*

- Applied and Computational Mechanics, 2006. Berlin, Heidelberg: Springer; 2006. p. 297–304. https://doi.org/10.1007/3-540-31761-9_33.
- [47] Sapora A, Paggi M. A coupled cohesive zone model for transient analysis of thermoelastic interface debonding. *Comput Mech* 2014;53:845–57. <https://doi.org/10.1007/s00466-013-0934-8>.
- [48] Hübner S, Wohlmuth BI. Mortar methods for contact problems. In: Wriggers P, Nackenhorst U, editors. *Analysis and Simulation of Contact Problems*. Lecture Notes in Applied and Computational Mechanics, 2006. Berlin, Heidelberg: Springer; 2006. p. 39–47. https://doi.org/10.1007/3-540-31761-9_5.
- [49] Belgacem FB, Hild P, Laborde P. The mortar finite element method for contact problems. *Math Comput Model* 1998;28:263–71. [https://doi.org/10.1016/S0895-7177\(98\)00121-6](https://doi.org/10.1016/S0895-7177(98)00121-6).
- [50] Puso MA, Laursen TA. A mortar segment-to-segment contact method for large deformation solid mechanics. *Comput Methods Appl Mech Eng* 2004;193:601–29. <https://doi.org/10.1016/j.cma.2003.10.010>.
- [51] Sahu I, Petrinic N. Patch based unbiased 3D frictional contact formulation for finite element algorithms. *Comput Methods Appl Mech Eng* 2025;441:117958. <https://doi.org/10.1016/j.cma.2025.117958>.
- [52] Akula B.R., Vignollet J., Yastrebov V.A. MorteX method for contact along real and embedded surfaces: coupling X-FEM with the Mortar method. *ArXiv Preprint arXiv:1902.04000* 2019. doi:10.48550/arXiv.1902.04000.
- [53] Yang B, Laursen TA, Meng X. Two dimensional mortar contact methods for large deformation frictional sliding. *Int J Numer Methods Eng* 2005;62:1183–225. <https://doi.org/10.1002/nme.1222>.
- [54] Wriggers P, Simo JC. A note on tangent stiffness for fully nonlinear contact problems. *Commun Numer Methods Eng* 1985;1:199–203. <https://doi.org/10.1002/cnm.1630010503>.
- [55] Fang R, Farah P, Popp A, Wall WA. A monolithic, mortar-based interface coupling and solution scheme for finite element simulations of lithium-ion cells. *Int J Numer Methods Eng* 2018;114:1411–37. <https://doi.org/10.1002/nme.5792>.
- [56] Williams ML. Stress singularities resulting from various boundary conditions in angular corners of plates in extension. *J Appl Mech* 1952;19:526–8. <https://doi.org/10.1115/1.4010553>.
- [57] Hills DA, Fleury RMN, Dini D. Partial slip incomplete contacts under constant normal load and subject to periodic loading. *Int J Mech Sci* 2016;108–109:115–21. <https://doi.org/10.1016/j.ijmecsci.2016.01.034>.
- [58] Hills DA, Moore MR. Partial slip problems in contact mechanics. Elsevier Inc.; 2022. <https://doi.org/10.1016/B978-0-12-824096-0.00012-3>.
- [59] Hills DA, Thaitirarat A, Barber JR, Dini D. Correlation of fretting fatigue experimental results using an asymptotic approach. *Int J Fatigue* 2012;43:62–75. <https://doi.org/10.1016/j.ijfatigue.2012.02.006>.
- [60] Hills DA, Nowell D, Sackfield A. *Mechanics of Elastic Contacts*. Oxford: Butterworth-Heinemann; 1993. <https://doi.org/10.1016/B978-0-7506-0540-3.50021-9>.
- [61] Hills DA, Andresen HN. *Mechanics of Fretting and Fretting Fatigue*. 1st ed. Cham: Springer Nature Switzerland AG; 2021. <https://doi.org/10.1007/978-3-030-70746-0>.
- [62] Hills DA, Dini D. A new method for the quantification of nucleation of fretting fatigue cracks using asymptotic contact solutions. *Tribol Int* 2006;39:1114–22. <https://doi.org/10.1016/j.triboint.2006.02.041>.
- [63] Korsunsky A.M. Power law multi-scaling of material strength. *ArXiv Preprint cond-mat/0508653* 2005. doi:10.48550/arXiv.cond-mat/0508653.
- [64] Liu J, Ma C, Wang S, Wang S, Yang B. Contact stiffness of spindle-tool holder based on fractal theory and multi-scale contact mechanics model. *Mech Syst Signal Process* 2019;119:363–79. <https://doi.org/10.1016/j.ymssp.2018.09.037>.
- [65] Zhai C, Gan Y, Hanaor D, Proust G, Retraint D. The Role of Surface Structure in Normal Contact Stiffness. *Exp Mech* 2016;56:359–68. <https://doi.org/10.1007/s11340-015-0107-0>.
- [66] Korsunsky AM, Constantinescu A. The influence of indenter bluntness on the apparent contact stiffness of thin coatings. *Thin Solid Films* 2009;517:4835–44. <https://doi.org/10.1016/j.tsf.2009.03.018>.
- [67] Fouvry S, Merhej R. Introduction of a power law formulation to quantify the contact size effects on friction and wear responses of dry oscillating sliding contacts: application to a chromium steel interface. *Wear* 2013;301:34–46. <https://doi.org/10.1016/j.wear.2013.01.072>.
- [68] Ciavarella M, Joe J, Papangelo A, Barber JR. The role of adhesion in contact mechanics. *J R Soc Interface* 2019;16. <https://doi.org/10.1098/rsif.2018.0738>.
- [69] Chen S, Yan C, Zhang P, Gao H. Mechanics of adhesive contact on a power-law graded elastic half-space. *J Mech Phys Solids* 2009;57:1437–48. <https://doi.org/10.1016/j.jmps.2009.06.006>.
- [70] Guo X, Jin F, Gao H. Mechanics of non-slipping adhesive contact on a power-law graded elastic half-space. *Int J Solids Struct* 2011;48:2565–75. <https://doi.org/10.1016/j.ijsolstr.2011.05.008>.
- [71] Li Q, Popov VL. Boundary element method for normal non-adhesive and adhesive contacts of power-law graded elastic materials. *Comput Mech* 2018;61:319–29. <https://doi.org/10.1007/s00466-017-1461-9>.
- [72] Johnson KL. *Contact mechanics*. Cambridge: Cambridge University Press; 1985. <https://doi.org/10.1017/CBO9781139171731>.